

A CANDIDATE PROOF OF THE REAL HILBERT SPACE FACTORIZATION CONJECTURE VIA ROTATING CONTACT CHORDS

EXPLORATORY SOLUTION REPORT

ABSTRACT. Aubrun and Müller-Hermes conjectured that a pair (X, Y) of real Banach spaces satisfies $\gamma_2(T) = \|T\|$ for every bounded operator $T : X \rightarrow Y$ if and only if one of X, Y is Hilbert. They reduced the question to an ellipse-sandwich problem for two centrally symmetric planar convex bodies.

The supplied automatic-solver result—a common quotient-subspace obstruction—is correct, but it addresses only a special configuration. A literature search through 22 June 2026 found no published proof or counterexample to the full conjecture. This report gives a complete candidate proof. The geometric argument puts the Löwner ellipse of the first body and the John ellipse of the second body in Euclidean position, sends their contact directions through the quadratic double-angle map, and rotates exposed contact chords so that they cross. A separating quadratic form then produces an arbitrarily small linear perturbation which is still a contraction but cannot admit an intermediate ellipse. The obstruction is certified by two probability measures with identical second moments.

This is a new, unrefereed argument developed in the present exploration.

PACKET STATUS AND SOURCE EVIDENCE

This is an upgraded full-solution packet for arXiv:2410.23063, Conjecture 5.2 of Aubrun–Müller-Hermes. The packet promotes the previous common quotient-subspace partial result to a candidate full proof of the real Hilbert space factorization property classification, via the equivalent two-dimensional ellipse-sandwich conjecture. The proof is presented as new and unrefereed, so the registry status is “full solution, likely valid” rather than externally verified.

- Previous active partial packet preserved in the dated history folder.
- Supplied TeX/PDF report preserved in the dated evidence folder.
- Source paper preserved as `source_paper.pdf`.

as needed. □

In the real case, we conjecture that there is no non-trivial pair with the HFP.

Conjecture 5.2. *A pair (X, Y) of real Banach spaces has the HFP if and only if X or Y is a Hilbert space.*

The pair $(\ell_\infty^2(\mathbf{C}), \ell_1^2(\mathbf{C}))$, when these spaces are seen as normed spaces over the reals, fails the HFP. This fact, which answers a question from [AMH24], can be seen using the following proposition.

Proposition 5.3. *If X and Y are non-strictly convex real Banach spaces, then the pair (X, Y) fails the HFP.*

It is a classical fact (see [Ami86]) that a normed space X is Euclidean if and only if all its 2-dimensional subspaces are Euclidean, and also if and only if all its 2-dimensional quotients are Euclidean. Using Proposition 5.4, this shows that in order to prove Conjecture 5.2 it suffices to consider the case $\dim X = \dim Y = 2$. This allows to formulate a concrete geometric question equivalent to Conjecture 5.2.

Conjecture 5.5. *Let K and L be centrally symmetric convex bodies in $\mathbf{R}^2$. Assume that for every linear map T such that $T(K) \subset L$, there is an ellipse $\mathcal{E} \subset \mathbf{R}^2$ such that $T(K) \subset \mathcal{E} \subset L$. Then K or L is an ellipse.*

1. THE QUESTION AND THE OUTCOME

For a bounded operator $T : X \rightarrow Y$, its Hilbert space factorization norm is

$$\gamma_2(T) = \inf\{\|a\| \|b\| : T = ba, a : X \rightarrow H, b : H \rightarrow Y, H \text{ Hilbert}\}.$$

A pair (X, Y) has the *Hilbert space factorization property* (HFP) if $\gamma_2(T) = \|T\|$ for every bounded $T : X \rightarrow Y$.

The source paper posed the following real classification problem [1, Conjecture 5.2].

Theorem 1.1 (Real HFP classification). *Let X, Y be real Banach spaces. The pair (X, Y) has the HFP if and only if X or Y is a Hilbert space.*

The nontrivial implication follows from the planar result below. Throughout, an ellipse may be taken to be centered at the origin. This entails no loss: if $A = -A$, $L = -L$, and an affine ellipse E satisfies $A \subset E \subset L$, then

$$E_0 := \frac{1}{2}(E - E)$$

is a centered ellipse and $A \subset E_0 \subset L$.

Theorem 1.2 (Planar ellipse-sandwich theorem). *Let $K, L \subset \mathbb{R}^2$ be centrally symmetric convex bodies. Assume that for every linear map T satisfying $T(K) \subset L$, there is an ellipse $\mathcal{E}$ such that*

$$T(K) \subset \mathcal{E} \subset L.$$

Then K or L is an ellipse. More precisely, if neither body is an ellipse, there is an invertible linear map T with $T(K) \subset L$ for which no intermediate ellipse exists.

This is exactly the geometric statement formulated as Conjecture 5.5 in the source paper. Sections 4–7 prove Theorem 1.2; Section 8 then deduces Theorem 1.1.

2. ASSESSMENT OF THE SUPPLIED PARTIAL RESULT

The automatic solver proved the following conditional obstruction.

Proposition 2.1 (Common quotient–subspace obstruction). *Let X, Y be real Banach spaces. Suppose that a finite-dimensional, non-Hilbert Banach space E is linearly isometric both to a quotient of X and to a subspace of Y . Then (X, Y) fails the HFP.*

Proof. For an n -dimensional Banach space E ,

$$(1) \quad \gamma_2(\text{Id}_E) = d(E, \ell_2^n).$$

Indeed, if $\text{Id}_E = ba$ through a Hilbert space, then a is injective and $b|_{a(E)} = a^{-1}$, whence

$$\|a\| \|b\| \geq \|a\| \|a^{-1} : a(E) \rightarrow E\| \geq d(E, \ell_2^n).$$

Conversely, every isomorphism $S : E \rightarrow \ell_2^n$ gives $\text{Id}_E = S^{-1}S$. This proves (1). In finite dimension the Banach–Mazur infimum is attained, and distance one means that the unit ball is an ellipsoid; hence $d(E, \ell_2^n) > 1$ when E is not Hilbert.

Aubrun and Müller-Hermes proved that HFP descends to quotients of the domain and subspaces of the range [1, Proposition 5.4]. Thus HFP for (X, Y) would imply HFP for (E, E) , contradicting

$$1 = \|\text{Id}_E\| < \gamma_2(\text{Id}_E).$$

□

The proof is correct as written. In particular, there is no hidden approximation issue in (1). Its limitation is structural: the same non-Euclidean space must occur isometrically on both sides. The planar corollary in the supplied packet, concerning linearly equivalent K and L , is also correct (and follows immediately by applying the sandwich property to a map with $T(K) = L$). Theorem 1.2 removes this common-space hypothesis entirely.

3. LITERATURE STATUS BEFORE THE NEW ARGUMENT

The conjecture appears in the 2024 preprint and in the 2025 Journal of Functional Analysis version of *Limit formulas for norms of tensor power operators* [1]. That paper proves, among other partial results, that two non-strictly convex real spaces cannot form an HFP pair and reduces the general real problem to dimension two.

The preceding paper on asymptotic tensor powers gives additional obstructions: a mismatch between John and Löwner contact points, the classification for real $\ell_p^m - \ell_q^n$ pairs, and several unresolved smooth examples [2, Section 5.2]. Searches by the exact HFP terminology, the two conjecture numbers, the article title and DOI, author publication lists, and forward citations revealed no later proof or counterexample through 22 June 2026. The only indexed forward citations found were the earlier asymptotic-tensor paper (whose final publication cross-references the follow-up) and work on Lorentz-cone entanglement; neither resolves the conjecture.

Accordingly, the proof below should be viewed as a new candidate resolution, not as a report of a known theorem. All steps needed for verification are included.

4. JOHN–LÖWNER NORMALIZATION AND CONTACT MOMENTS

For a compact convex body $C \subset \mathbb{R}^2$, write

$$h_C(u) = \sup_{x \in C} \langle u, x \rangle$$

for its support function. Let B_2 be the Euclidean unit disk and $\mathbb{S}^1 = \partial B_2$.

The sandwich property is invariant under independent invertible changes of coordinates in K and L . We may therefore send the Löwner ellipse of K (the minimum-area ellipse containing K) to B_2 , and independently send the John ellipse of L (the maximum-area ellipse contained in L) to B_2 . Thus from now on

$$(2) \quad K \subset B_2 \subset L, \quad \mathcal{L}(K) = B_2, \quad \mathcal{J}(L) = B_2.$$

Define the two Euclidean contact sets

$$\begin{aligned} C_K &:= K \cap \mathbb{S}^1, \\ C_L &:= \{u \in \mathbb{S}^1 : h_L(u) = 1\}. \end{aligned}$$

Because $B_2 \subset L$, the second set is also $\partial L \cap \mathbb{S}^1$. Indeed, a common boundary point of L and the strictly convex disk has the same supporting line.

We use the quadratic double-angle map

$$(3) \quad \nu : \mathbb{S}^1 \longrightarrow \mathbb{S}^1, \quad \nu(u_1, u_2) = (u_1^2 - u_2^2, 2u_1u_2).$$

It identifies u and $-u$ and satisfies

$$(4) \quad uu^\top = \frac{1}{2} \begin{pmatrix} 1 + \nu_1(u) & \nu_2(u) \\ \nu_2(u) & 1 - \nu_1(u) \end{pmatrix}.$$

A physical rotation through angle θ becomes a rotation through angle 2θ after applying ν .

Lemma 4.1 (Isotropic contact images). *Under (2), the closed sets*

$$A := \nu(C_K), \quad B := \nu(C_L)$$

satisfy

$$0 \in \text{conv } A \cap \text{conv } B.$$

Moreover, $A = \mathbb{S}^1$ if and only if $K = B_2$, and $B = \mathbb{S}^1$ if and only if $L = B_2$.

Proof. The symmetric John–Löwner theorem gives contact points $u_i \in C_K$ and positive weights c_i such that [5]

$$\sum_i c_i u_i u_i^\top = I_2.$$

Taking traces gives $\sum_i c_i = 2$. Formula (4) then gives $\sum_i (c_i/2) \nu(u_i) = 0$, so $0 \in \text{conv } A$. The John decomposition for $B_2 \subset L$ gives the same conclusion for B .

If $A = \mathbb{S}^1$, central symmetry of C_K implies $C_K = \mathbb{S}^1$, hence $B_2 \subset K$ and therefore $K = B_2$. The converse is immediate. The argument for L is analogous, using $h_L(u) = 1$ in every direction. $\square$

The key observation is that equal barycenters after the map ν are exactly equal second moments before the map.

Lemma 4.2 (Moment lifting). *Let μ, η be probability measures on $\mathbb{S}^1$. If*

$$\int \nu(u) d\mu(u) = \int \nu(v) d\eta(v),$$

then

$$\int uu^\top d\mu(u) = \int vv^\top d\eta(v).$$

Proof. Integrate (4). $\square$

5. A CIRCLE LEMMA: ROTATE TWO MISSING CONTACT ARCS UNTIL THEIR CHORDS CROSS

The following elementary lemma is the topological-geometric core of the proof.

Lemma 5.1 (Crossing contact chords). *Let $A, B \subset \mathbb{S}^1$ be proper closed subsets satisfying $0 \in \text{conv } A \cap \text{conv } B$. Then there are a rotation ρ of $\mathbb{R}^2$ and a point p such that*

$$(5) \quad p \in \text{conv}(\rho A) \cap \text{conv } B, \quad p \notin \text{conv}(\rho A \cap B).$$

Here the convex hull of the empty set is understood as empty.

Proof. Since A is proper and closed, $\mathbb{S}^1 \setminus A$ has a nonempty open arc component I_A . Its angular length is at most π : otherwise A would be contained in an open semicircle, contrary to $0 \in \text{conv } A$. Let a_-, a_+ be the endpoints. The chord

$$F_A = [a_-, a_+]$$

is an exposed face of $\text{conv } A$. More explicitly, there are a unit normal m_A and a number $r_A \in [0, 1)$ such that

$$\text{conv } A \subset \{x : \langle m_A, x \rangle \leq r_A\}, \quad F_A = \text{conv } A \cap \{x : \langle m_A, x \rangle = r_A\}.$$

The number r_A is the distance of the chord line from the origin. Choose similarly an exposed gap chord F_B with unit normal m_B and distance $r_B \in [0, 1)$.

We can rotate F_A so that the relative interiors of F_A and F_B cross. For completeness, suppose first that $r_A > r_B$. Rotate its normal to a unit vector m satisfying

$$\langle m_B, m \rangle = r_B / r_A.$$

Then $p = r_A m$ is the midpoint of the rotated F_A , lies on the line of F_B , and has $|p| = r_A < 1$; hence it lies in the relative interior of both chords. The case $r_B > r_A$ is symmetric. If $r_A = r_B = r > 0$, take two nonparallel normals at a sufficiently small angle ϕ ; their line intersection has norm $r / \cos(\phi/2) < 1$. If both distances are zero, use two nonparallel diameters, which meet at the origin. Let ρ be the resulting rotation of A .

The crossing point p belongs to both convex hulls. Suppose that $p \in \text{conv}(\rho A \cap B)$. In a convex representation of p by common points, the supporting functional of the rotated F_A has average equal to its maximum r_A , so every point used must lie on F_A . The same argument with F_B forces every point used to lie on F_B . The two supporting lines are nonparallel and meet only at p , while $|p| < 1$ and all common points lie on $\mathbb{S}^1$. This is impossible, proving (5). $\square$

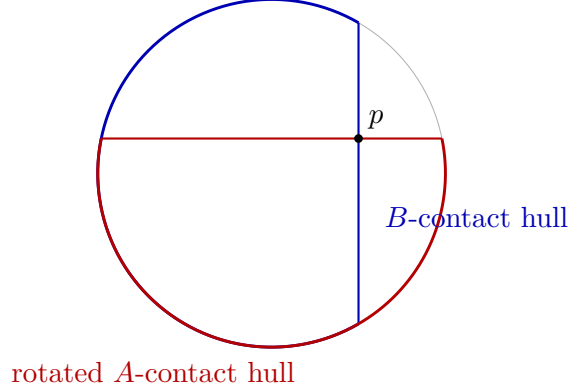


FIGURE 1. In the doubled-angle circle, a gap in each proper contact set creates an exposed chord. One chord can be rotated so that the two relative interiors cross. The crossing point belongs to both contact hulls but not to the hull of the common contact directions.

6. THE PERTURBATION AND THE EQUAL-MOMENT OBSTRUCTION

We now convert Lemma 5.1 into a linear contraction with no ellipse sandwich.

Lemma 6.1 (Perturbation obstruction). *Assume (2). Let R be an orthogonal map. Suppose there are probability measures μ on RC_K and η on C_L such that*

$$(6) \quad M := \int uu^\top d\mu(u) = \int vv^\top d\eta(v),$$

and a symmetric matrix H such that

$$(7) \quad \langle Hu, u \rangle < 0 \quad \text{for every } u \in RC_K \cap C_L,$$

$$(8) \quad \int \langle Hu, u \rangle d\mu(u) > 0.$$

Then, for every sufficiently small $t > 0$, the map

$$T_t = (I_2 + tH)R$$

satisfies $T_t(K) \subset L$, but there is no ellipse $\mathcal{E}$ with $T_t(K) \subset \mathcal{E} \subset L$.

Proof. We first prove $T_t(K) \subset L$. If not for arbitrarily small t , there would be $t_n \downarrow 0$, $x_n \in K$, and $u_n \in \mathbb{S}^1$ such that

$$(9) \quad \langle u_n, (I_2 + t_n H)Rx_n \rangle > h_L(u_n).$$

After taking a subsequence, let $x_n \rightarrow x$ and $u_n \rightarrow u$. At $t = 0$,

$$\langle u_n, Rx_n \rangle \leq h_{RK}(u_n) \leq 1 \leq h_L(u_n).$$

Consequently (9) and $t_n \rightarrow 0$ force equality throughout in the limit. Equality in the Euclidean Cauchy inequality gives $Rx = u$; also $u \in RC_K$ and $h_L(u) = 1$, so $u \in RC_K \cap C_L$. On the other hand,

$$0 < \langle u_n, (I_2 + t_n H)Rx_n \rangle - h_L(u_n) \leq t_n \langle u_n, H Rx_n \rangle.$$

The last coefficient converges to $\langle Hu, u \rangle < 0$, a contradiction. This proves the inclusion for all sufficiently small t .

Suppose now that a centered ellipse $\mathcal{E}$ satisfies the sandwich. Write

$$h_{\mathcal{E}}(w)^2 = w^\top Q w$$

for a positive definite matrix Q . For $u \in RC_K$, the point $R^{-1}u$ belongs to K , so

$$(10) \quad h_{\mathcal{E}}(u) \geq h_{T_t K}(u) \geq \langle u, (I_2 + tH)u \rangle = 1 + t\langle Hu, u \rangle.$$

For small t the right side is positive. For $v \in C_L$, the upper inclusion gives

$$(11) \quad v^T Q v = h_{\mathcal{E}}(v)^2 \leq h_L(v)^2 = 1.$$

Using the common moment matrix (6), we obtain

$$\begin{aligned} 1 &\geq \int v^T Q v d\eta(v) = \text{tr}(QM) = \int u^T Q u d\mu(u) \\ &\geq \int (1 + t\langle Hu, u \rangle)^2 d\mu(u) \\ &= 1 + 2t \int \langle Hu, u \rangle d\mu(u) + t^2 \int \langle Hu, u \rangle^2 d\mu(u) > 1, \end{aligned}$$

where the last inequality follows from (8). This contradiction excludes every intermediate ellipse. $\square$

The hypotheses of Lemma 6.1 follow automatically from a point p as in Lemma 5.1.

Lemma 6.2 (Separation produces the perturbation). *Let $A = \nu(C_K)$ and $B = \nu(C_L)$. Suppose a rotation ρ and a point p satisfy (5). Let R be a physical rotation inducing ρ under ν . Then there are μ, η, H satisfying (6)–(8).*

Proof. Represent p by probability measures on the two contact sets:

$$p = \int \nu(u) d\mu(u), \quad \text{supp } \mu \subset RC_K,$$

$$p = \int \nu(v) d\eta(v), \quad \text{supp } \eta \subset C_L.$$

In fact, the two endpoints of each crossing chord suffice. By Lemma 4.2, these measures have the same second moment.

Central symmetry and the fact that $\nu(u) = \nu(v)$ exactly when $v = \pm u$ give

$$\nu(RC_K \cap C_L) = \rho A \cap B.$$

If $RC_K \cap C_L$ is nonempty, strict separation of p from the compact convex set

$$\text{conv } \nu(RC_K \cap C_L) = \text{conv}(\rho A \cap B)$$

gives an affine functional

$$\ell(z) = \alpha + \beta_1 z_1 + \beta_2 z_2$$

which is strictly negative on that set and satisfies $\ell(p) > 0$. If the common contact set is empty, take simply $\ell \equiv 1$. Define

$$H = \begin{pmatrix} \alpha + \beta_1 & \beta_2 \\ \beta_2 & \alpha - \beta_1 \end{pmatrix}.$$

Then, by (3),

$$\langle Hu, u \rangle = \ell(\nu(u)).$$

It is strictly negative on $RC_K \cap C_L$, while

$$\int \langle Hu, u \rangle d\mu(u) = \ell\left(\int \nu(u) d\mu(u)\right) = \ell(p) > 0.$$

$\square$

7. PROOF OF THE PLANAR THEOREM

Proof of Theorem 1.2. Normalize as in (2). Suppose for contradiction that neither K nor L is an ellipse. Then $K \neq B_2$ and $L \neq B_2$. By Lemma 4.1, the doubled contact sets

$$A = \nu(C_K), \quad B = \nu(C_L)$$

are proper closed subsets of the circle and both have the origin in their convex hulls. Lemma 5.1 supplies a doubled rotation ρ and a point p satisfying (5). Choose the corresponding physical rotation R . Lemma 6.2 gives equal-moment measures and a symmetric matrix H , and Lemma 6.1 gives, for small $t > 0$, an invertible linear map T_t such that

$$T_t(K) \subset L$$

but no ellipse lies between $T_t(K)$ and L . This contradicts the assumed sandwich property. Hence K or L is an ellipse. $\square$

Remark 7.1 (Constructive content). The proof gives an explicit finite recipe for a bad operator. Choose one missing arc in each doubled contact set, rotate their exposed chords to cross, write the crossing point as a convex combination of each chord's endpoints, and separate it from the convex hull of the common contacts. The separating affine functional determines H , and every sufficiently small $(I_2 + tH)R$ is a witness. Thus the proof is not merely existential at the level of operator geometry.

8. DEDUCTION OF THE BANACH-SPACE CLASSIFICATION

We first record the exact factorization–ellipse dictionary needed below.

Lemma 8.1 (Factorizations and ellipse sandwiches). *Let K, L be the unit balls of two real two-dimensional normed spaces, and let T be invertible. Then*

$$\gamma_2(T) \leq 1$$

if and only if there is a centered ellipse $\mathcal{E}$ such that

$$T(K) \subset \mathcal{E} \subset L.$$

Proof. Suppose first that $\mathcal{E} = b(B_2)$, where $b : \ell_2^2 \rightarrow \mathbb{R}^2$ is invertible. The two inclusions say that

$$\|b^{-1}T : K \rightarrow B_2\| \leq 1, \quad \|b : B_2 \rightarrow L\| \leq 1,$$

so $T = b(b^{-1}T)$ is a Hilbert factorization of product at most one.

Conversely, in finite dimensions the infimum defining $\gamma_2(T)$ is attained through ℓ_2^2 . Indeed, in every approximate factorization $T = ba$ one may restrict the Hilbert space to $a(\mathbb{R}^2)$, identify that 2-dimensional Hilbert space with ℓ_2^2 , and rescale so that $\|a\| = \|b\|$; compactness of bounded sets of matrices then gives a minimizing pair. Thus write $T = ba$ with $\|a\| \|b\| \leq 1$. Rescaling the two factors once more, we may arrange $\|a\| \leq 1$ and $\|b\| \leq 1$. Since T is invertible, so is b , and

$$\mathcal{E} := b(B_2)$$

is an ellipse satisfying $T(K) \subset \mathcal{E} \subset L$. $\square$

The source paper states this dictionary as the equivalence between its Conjectures 5.2 and 5.5. Lemma 8.1 also shows directly why the invertibility supplied by Theorem 1.2 eliminates any issue about degenerate ellipses or non-attainment.

Proof of Theorem 1.1. If X is Hilbert, then every $T : X \rightarrow Y$ factors through X as $T = T \text{Id}_X$, so $\gamma_2(T) \leq \|T\|$; the reverse inequality always holds. The same argument applies when Y is Hilbert.

Conversely, suppose (X, Y) has the HFP and neither space is Hilbert. A classical characterization of inner-product spaces says that a real Banach space is Hilbert if and only if all of its two-dimensional subspaces are Euclidean, and also if and only if all of its two-dimensional quotients are Euclidean

[3]. Hence X has a non-Euclidean two-dimensional quotient X/F , and Y has a non-Euclidean two-dimensional subspace E . HFP descends to $(X/F, E)$ by [1, Proposition 5.4].

Let K, L be the unit balls of X/F and E . Neither is an ellipse, so the more precise conclusion of Theorem 1.2 gives an invertible map T with $T(K) \subset L$ and with no intermediate ellipse. On the other hand, HFP gives

$$\gamma_2(T) = \|T\| \leq 1,$$

and Lemma 8.1 produces exactly such an ellipse, a contradiction. Therefore X or Y is Hilbert. $\square$

9. RELATION TO THE ORIGINAL PARTIAL RESULT AND PRIOR OBSTRUCTIONS

The common quotient–subspace argument in Proposition 2.1 is a valid special case and remains a useful one-line certificate. The full proof uses a different mechanism. Rather than seeking a common non-Hilbert space, it extracts two *different* contact measures with the same Euclidean second moment. Any intermediate ellipse evaluates those measures identically, whereas the carefully chosen perturbation makes the lower evaluation strictly larger than the upper one.

The contact-chord step also explains why earlier generic-contact arguments do not cover bodies with large contact arcs. Even when every rotated pair of contact sets intersects, properness creates a missing arc in each doubled contact set. Their exposed chords can be made to cross in the interior of the disk, and that interior crossing—not absence of common contacts—is the required obstruction. In particular, it covers the smooth or nearly Euclidean examples left open in [2, Section 5.2].

10. VERIFICATION CHECKLIST AND STATUS

The proof rests on four exact ingredients:

- (i) the symmetric John–Löwner contact decomposition;
- (ii) the elementary identity (4);
- (iii) the crossing-chord Lemma 5.1;
- (iv) the integrated ellipse inequalities in Lemma 6.1.

No numerical optimization or regularity assumption on K, L is used. The bodies may be nonsmooth and non-strictly convex. The only limiting argument needed for the Banach formulation is finite-dimensional compactness of ellipses; the central geometric contradiction itself is quantitative for all sufficiently small $t > 0$.

The supplied partial proof has been verified. The literature search did not locate an existing full solution. Theorem 1.2 and its deduction of Theorem 1.1 are therefore presented as a new proof candidate and should be checked independently before citation or submission.

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